

The anti-echinococcal activity of crocin in mice experimentally infected with *Echinococcus granulosus*

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ABSTRACT

Herbal preparations have good medicinal value for potential use as therapeutic agents in cystic echinococcosis. The efficiency of crocin in the case of cystic echinococcosis was investigated, and compared with that of albendazole, one of the few licensed anti-echinococcosis drugs that served as a positive control.

Five months after infecting BALB/C mice with *E. granulosus* sensu lato the experimental group was divided into 7 subgroups containing 10 animals each: 1- Crocin 80 (80 mg/kg), 2- Crocin 40 (40 mg/kg), 3- Crocin 10 (10 mg/kg), 4- Albendazole (200 mg/kg), 5- Crocin 10 (10 mg/kg) + Albendazole (100 mg/kg), 6- Crocin 20 (20 mg/kg) + Albendazole (50 mg/kg), 7- the control (infected and untreated).

After two weeks of daily treatment, significant reductions of cysts' weights, sizes, and total numbers concerning the control group were achieved by treatment with crocin 80, crocin 40, crocin 10, crocin 10 + ABZ100, crocin 20 + ABZ50 and ABZ200 ($p < 0.05$). Moreover, there was no difference concentrations of crocin and those treated with albendazole. The concentration of bilirubin was higher in the control group than all treated groups with crocin, significantly. However, the ALT activity showed a significant decrease in the crocin 10 group, compared to the crocin 80, crocin 40, crocin 10 + ABZ100, crocin 20 + ABZ50, control groups ($p < 0.05$).

Based on our results, the administration of crocin used at 10 mg/kg concentrations seems a hopeful applicant for the treatment of cystic echinococcosis.

1. Introduction

Cystic echinococcosis (CE) is a cosmopolitan zoonotic cestode disease in humans typically caused by infections from the oncospheres of *Echinococcus granulosus*. Currently, anti-parasitic treatment remains the only option for echinococcosis patients for whom surgery is contraindicated (Moro and Schantz, 2009). Moreover, even in individuals who undergo surgery, anti-parasitic treatment is always used to relieve or reduce the possible recurrence of this disease, which is the most common complication after radical resection of echinococcus masses (Moro and Schantz, 2009; Eckert and Deplazes, 2004). However, in clinical practice, anti-parasitic treatment is unsatisfactory due to many factors, such as the poor solubility of the medications and weak intestinal absorption. (El-On, 2003).

Herbal medicine proposes a substitute to overcome some of these complications, and they would be, both maintainable and biologically suitable. Therefore, a thorough understanding of the reactions performed by medicinal plants may provide essential insights into the

identification novel bioactive components from natural resources which would help to design efficient pharmacotherapies with minimal side effects. Several studies have identified new compounds of medicinal plants with activities against CE (Haji Mohammadi et al., 2018; Moazeni et al., 2014, 2017).

Saffron, a plant without stems which is in the family of Iridaceae, is utilized primarily as a spicing agent and food coloring. The pharmacologically active saffron contains three primary metabolites, including crocin, picrocrocin, safranal which is producing the color, the odor and the bitter savor of saffron, respectively (José Bagur et al., 2017; Christodoulou et al., 2015). Therapeutic effects of saffron have been reported in various disorders such as asthma, autoimmune, diabetes and cardiovascular, (Christodoulou et al., 2015; Khorasanchi et al., 2018). Of the carotenoids present in saffron, highly water-soluble crocin (mono and diglycosyl esters of a polyene dicarboxylic acid, named crocetin) is responsible for the majority of its color, and appears to possess various health-promoting properties, as an antioxidant, antitumor, memory enhancer, antidepressant, anxiolytic and aphrodisiac. It is also worth

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Table 1

The effect of different concentrations of Crocin on the treatment of cystic echinococcosis in experimentally infected BALB/c mice.

Group	Total number of cysts	Mean cysts weight (mg)	Mean macrocysts Size (mm)	Mean microcysts Size (mm)	Bilirubin (mg/dL)	ALT (IU/L)
Crocin 80	5.5 ± 2.07 ^a	0.4 ± 0.25 ^a	5.83 ± 1.16 ^a	2.5 ± 1.04 ^a	0.29 ± 0.02 ^a	65.33 ± 8.75
Crocin 40	3.29 ± 1.11 ^a	0.35 ± 0.12 ^a	8.14 ± 1.77 ^a	3.14 ± 0.69	0.24 ± 0.00 ^a	65.28 ± 7.31
Crocin 10	2 ± 1.34 ^a	0.65 ± 0.39 ^a	8.14 ± 2.79 ^a	3.86 ± 0.69	0.24 ± 0.02 ^a	58.41 ± 3.09 ^a
Albendazol 200	9 ± 3.13 ^a	1.18 ± 0.43 ^a	7.43 ± 1.71 ^a	2.29 ± 0.75 ^a	0.44 ± 0.00	82.71 ± 7.56
Crocin 10+ Albendazol 100	7 ± 2.61 ^a	0.28 ± 0.12 ^a	5.14 ± 2.11 ^a	2.29 ± 0.75 ^a	0.25 ± 0.01 ^a	71 ± 10.83
Crocin 20+ Albendazol 50	2.71 ± 0.75 ^a	0.7 ± 0.12 ^a	7 ± 2.82 ^a	1.86 ± 3.71 ^a	0.26 ± 0.02 ^a	65.57 ± 6.9
Control	12 ± 3.09	2.61 ± 1.59	14.29 ± 6.1	5.14 ± 1.58	0.49 ± 0.06	94.21 ± 10.28

a: significant difference with control.

_The smallest cyst as a microcyst and the largest cyst as a macrocyst.

noting that the crocin principle of saffron exhibited high efficacy along with no major toxicity in experimental models. (Christodoulou et al., 2015; Khorasanchi et al., 2018).

The potential antimalarial and anti-Leishmanial effects of two main compounds in saffron including crocin and safranal, were previously investigated (De Monte et al., 2015; Yousefi et al., 2014). However, there is no published data on the *in vivo* efficacy of this herb on *E. granulosus sensu lato*. Consequently, the objective of the current investigation was to compare the effect of crocin with that of albendazole in experimentally *E. granulosus sensu lato* infected mice and assess the effect of treatment on the liver function.

2. Materials and methods

2.1. Preparation of crocin

Pure reference standard of crocin from Sigma-Aldrich (St. Louis, Missouri, USA) was utilized for this study.

2.2. Collection of protoscoleces

Protoscoleces were aseptically isolated from cystic echinococcosis collected from naturally infected sheep from Mashhad Slaughterhouse in Northeast of Iran. For the confirmation of the protoscoleces viability, 0.1% eosin staining was used and then, the protoscoleces were washed several times with sterile 0.85% NaCl. Finally, they were stored at 4°C for injection.

2.3. Infection of mice

Seventy specific pathogen-free female BALB/c mice (approximately 24 ± 2 g) were purchased from the Razi Veterinary Research Institute, Mashhad, Iran. The animals were maintained in a controlled temperature (25 ± 2°C), light-cycled room, with water and food providing *ad libitum*.

All of BALB/c mice were infected by intraperitoneal injection with 500 µL of *E. granulosus sensu lato* metacestode-mass homogenates containing approximately 1500 protoscoleces.

2.4. Therapeutic trials

After the five months infection, the experimental group was divided into 7 subgroups containing 10 animals each: 1- Crocin 80 (80 mg/kg), 2- Crocin 40 (40 mg/kg), 3- Crocin 10 (10 mg/kg), 4- Albendazole (200 mg/kg), 5- Crocin 10 (10 mg/kg) +Albendazole (100 mg/kg), 6- Crocin 20 (20 mg/kg) +Albendazole (50 mg/kg), 7- the control (infected untreated). The mice were orally treated with crocin and, or albendazole by gavage with drinking water. Since there were no previous *in vitro* studies on the anti-parasitic effects of saffron on cystic echinococcosis, concentrations of crocin was determined based on previous articles from other parasites (Yousefi et al., 2014). In this study, oral suspension of albendazole 2.5% (Rooyan company, Iran) was used and the required

amount of this drug was dissolved in drinking water.

Blood samples were obtained after anesthesia from all the mice in each of the experimental groups after two weeks of treatment and were kept into glass tubes without anti-coagulant for biochemical analysis before euthanasia. For separation of serum, centrifugation at × 2000 g for 15 min was used and after that, the serum was stored at −20 °C until analysis.

2.5. Parasitological examinations

Twenty-four hours after the last treatment period (2 weeks), all the mice were euthanized, and necropsied immediately based on the AVMA Guidelines. During each necropsy, the cystic echinococcosis were carefully isolated and a digital scale (Kern&Sohn GmbH, Balingen, Germany) was used for weighing to evaluate the treatment effect of crocin, which was based on the total number of cysts, the weight and the size of the cyst. The size of all of the cysts were measured using a Scaled ruler and Adobe Photoshop CS6, which the smallest cyst considered as a microcyst and the largest cyst considered as a macrocyst in the table to evaluate the efficacy of the crocin.

2.6. Biochemical analysis

The activities of liver enzymes in the bloodstream from BALB/c mice were measured to evaluate liver function or liver injuries. Plasma levels of bilirubin and alanine aminotransferase (ALT) were used for the assessment of liver activity. These parameters were assessed using Pars Azmoon Lab (Iran) commercial kits utilizing an autoanalyzer (Biotecnica, Targa 3000, Rome, Italy). Enzymatic and bilirubin activities were exhibited in units per liter (U/L) and milligrams per deciliter (mg/dL).

2.7. Statistical analysis

SPSS 19.0 software was used for statistical analysis. The reported data were shown as mean ± standard deviation (SD). A one-way analysis of variance (ANOVA) was used for statistical comparisons. The results were considered statistically significant for *p* < 0.05.

2.8. Ethics

The animal trials were done under the basis of Good Laboratory Animal Care and were permitted with No.49002 by the Ethical Committee for Laboratory Animal Care and Use of the Ferdowsi University of Mashhad.

3. Results

3.1. Parasitological analysis

In the present study, the total number of cysts and the size and weight of the mass clusters of cyst collected from each mouse were

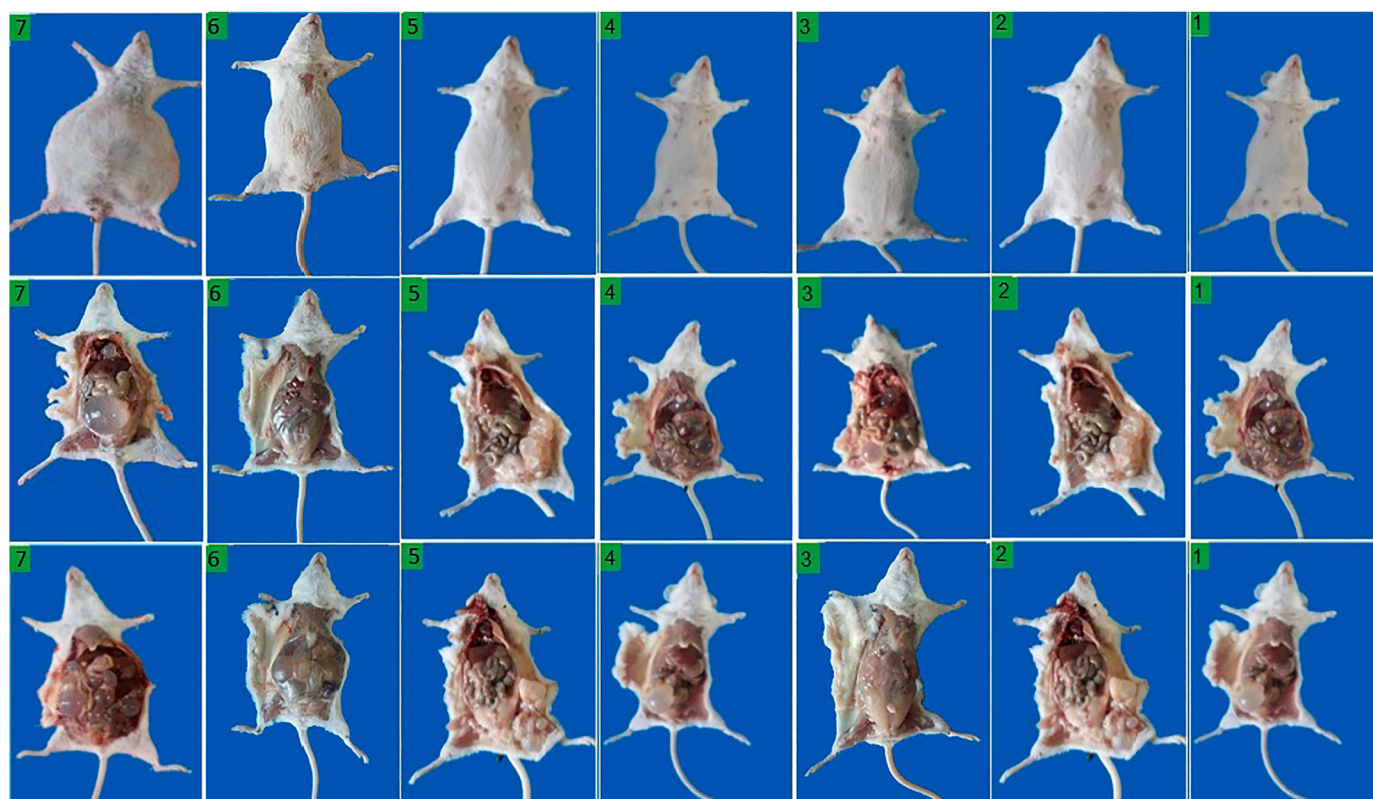


Fig. 1. Cystic echinococcosis in the peritoneal cavity and internal organs of the infected mice of the control and treated groups one month after treatment: 1- Crocin 80 (80 mg/kg), 2- Crocin 40 (40 mg/kg), 3- Crocin 10 (10 mg/kg), 4- Albendazole (200 mg/kg), 5- Crocin 10 (10 mg/kg) +Albendazole (100 mg/kg), 6- Crocin 20 (20 mg/kg) +Albendazole (50 mg/kg), 7- the control (infected and untreated).

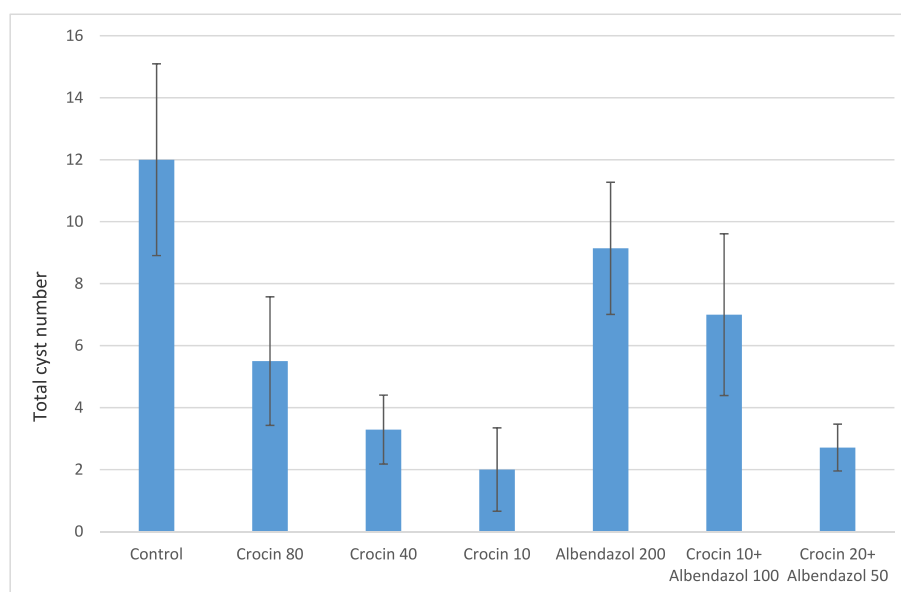


Fig. 2. Mean of total cyst number $\pm$ SD in both the treated and control groups.

compared between the different groups (Table 1, Figs. 1–5). Significant reductions of cyst weights, sizes, and total numbers concerning the control group were achieved by treating crocin 80, crocin 40, crocin 10, crocin 10 + ABZ100, crocin20+ ABZ50 and ABZ200 ($p < 0.05$). However, there were no significant difference for the total number of cysts and the size and weight of the mass clusters of cyst between the mice treated with albendazole and those treated with different concentrations

of crocin.

3.2. Biochemical analysis

Effects of crocin on bilirubin and ALT levels are shown in Figs. 6 and 7. In all treated groups with crocin compared to the control group, significantly lower concentrations of bilirubin were found. However, the

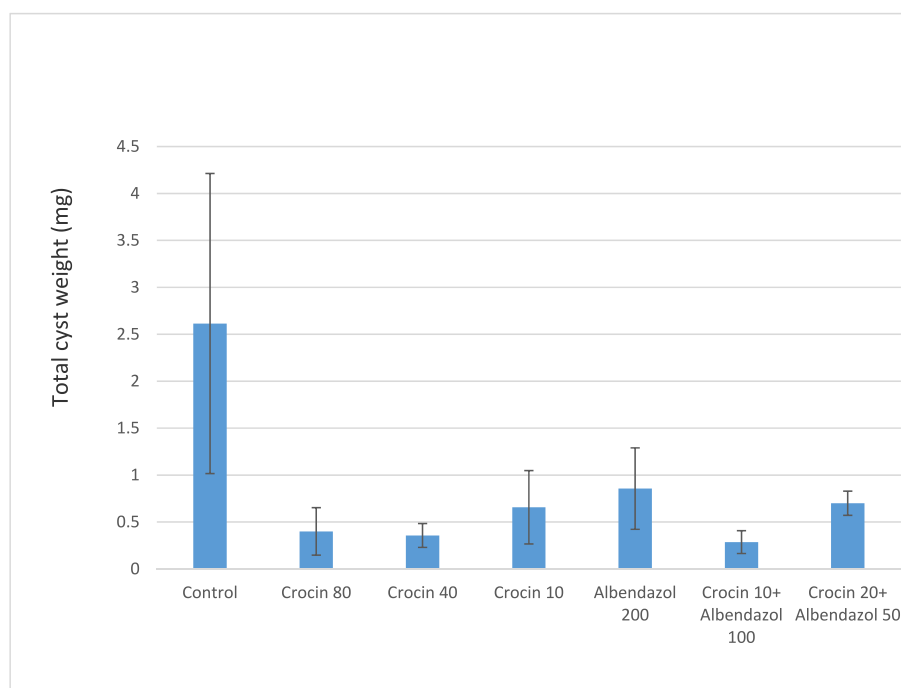


Fig. 3. Mean of total cyst weight (mg) $\pm$ SD in both the treated and control groups.

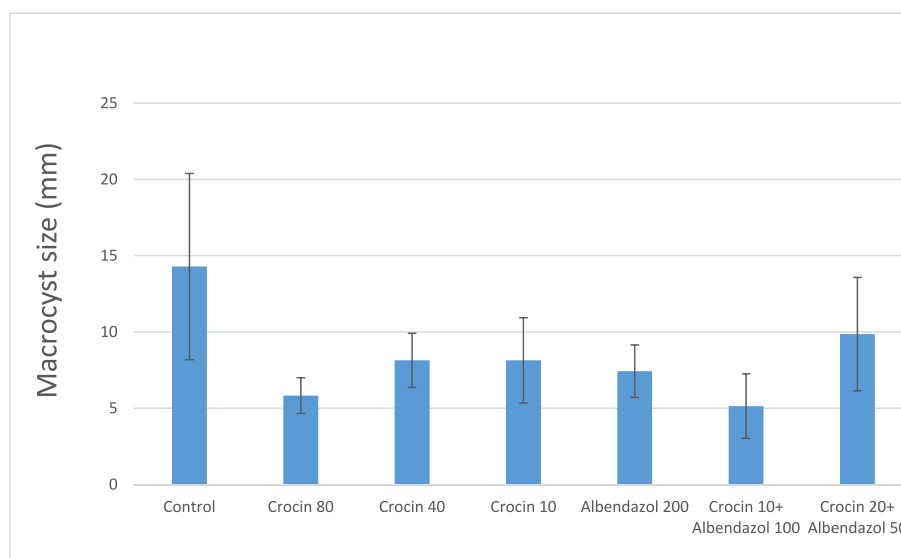


Fig. 4. Mean of macrocyst cyst (the largest cyst) size (mm) $\pm$ SD in both the treated and control groups.

ALT activity revealed a significant decrease in the crocin 10 group, compared to the crocin 80, crocin 40, crocin 10 + ABZ100, crocin 20 + ABZ50, and control groups ($p < 0.05$). There was no significant decrease in ALT activity after treatment with ABZ 200 as compared to the control group (Table 1).

4. Discussion

Herbal preparations have good medicinal value for potential use as therapeutic agents in CE. Therefore, the identification and validation of necessary information of plant-based anthelmintics will help develop rational treatments with few side effects as herbal medicine may have side-effects as well (Karimi et al., 2015). Therefore in this study, the efficiency of crocin against cystic echinococcosis was investigated and

compared with that of ABZ, one of the few licensed anti-echinococcosis drugs and served as a positive control.

The parasitological results revealed that crocin has high efficacy against *E. granulosus sensu lato* in mice as cyst mean weight was significantly reduced compared to the control group. This therapeutic result was consistent with that of albendazole (200 mg/kg/day). Moreover, our results revealed that the efficacy of crocin was not dose-dependent, and there is no significant difference in group crocin10 with crocin 80 and 40. However, in previous work on the treatment of CE with *Allium sativum* ME and *Zataria multiflora* revealed that the treatment with these two herbal medicine is dose-dependent (Moazeni et al., 2014; Yousefi et al., 2014). Our previous works on anti-echinococcal activity *A. sativum* ME and *Z. multiflora* indicated that higher concentrations resulted in more improved efficacy against CE in mice. The best

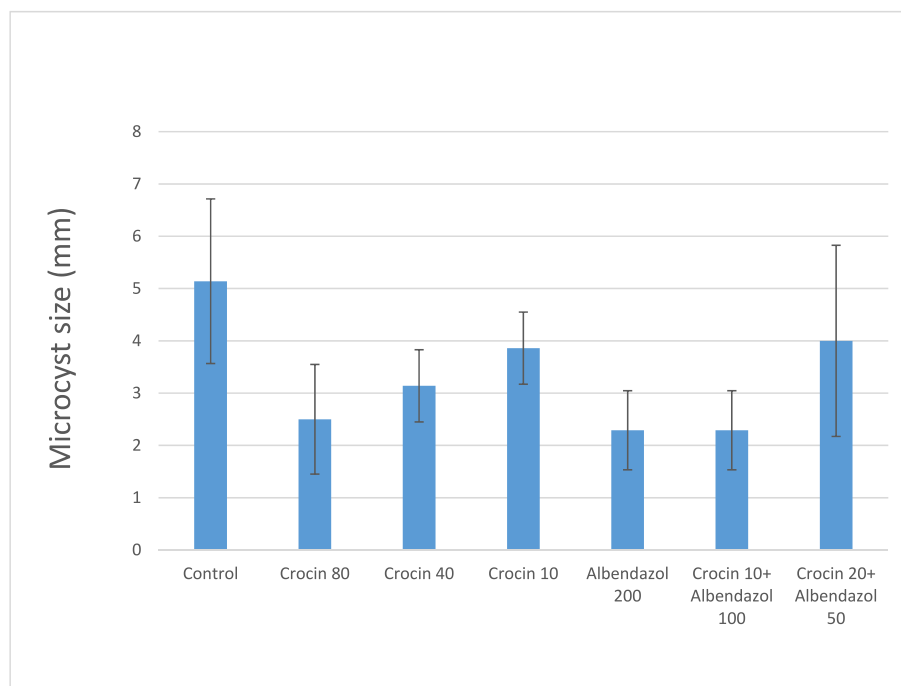


Fig. 5. Mean of microcyst cyst (the smallest cyst) size (mm) $\pm$ SD in both the treated and control.

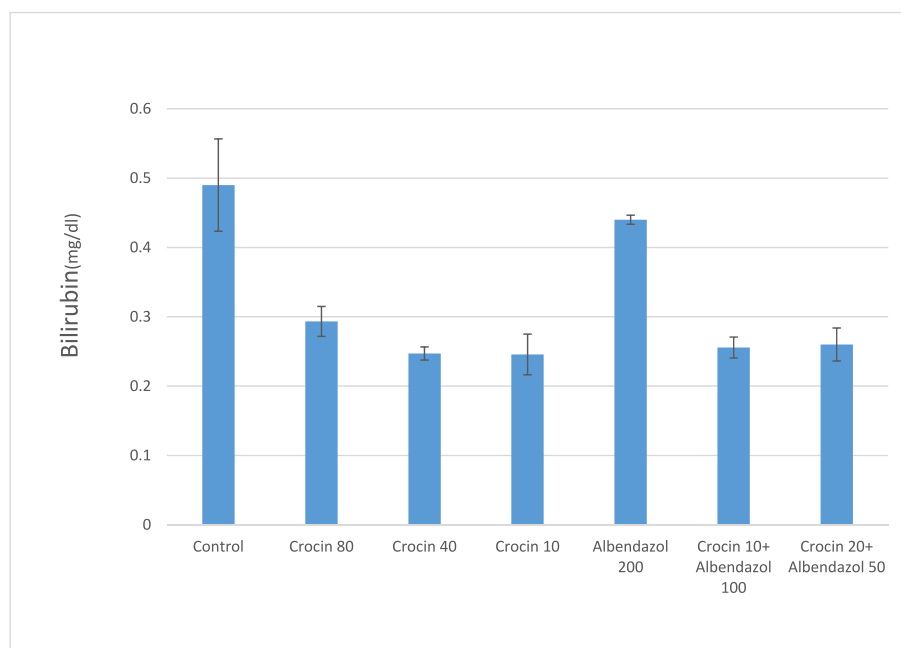


Fig. 6. Mean of bilirubin (mg/dL) $\pm$ SD in both the treated and control groups.

therapeutic findings based on the total number of cysts were obtained following the administration of *A. sativum* ME 80 and *Z. multiflora* 100 mL/L.

The mechanism of action of crocin, which is the main bioactive substance in saffron, against *E. granulosus* is not known. Several mechanisms such as the induction of apoptosis, the stunting cell cycle development, modulatory properties on various phase II clearing enzymes, suppressing the manifestation of matrix metalloproteinase, and reducing expression of inflammatory molecules, may be involved in antiparasite properties of saffron. Crocin suppressed intracellular ROS production and elevated total antioxidant activity, which reduced the development

of cystic echinococcosis (Zeinali et al., 2019). It is known that oxidative stress due to CE at the site of cyst formation and inflammation leads to the generation of lipid peroxidation (LPO) products, which may play a central role in the pathology of CE (Heidarpour et al., 2013).

Noticeably, saffron can stop chronic stress induced oxidative destruction of the liver (Ebrahimi et al., 2019). Consequently, these constituents should be valuable as novel pharmacological agent for ameliorating chronic stress induced oxidative damages, including hydatidosis hepatopathy.

Our results from the biochemical evaluation showed that plasma ALT levels were reduced in infected mice treated with low dose of crocin (10

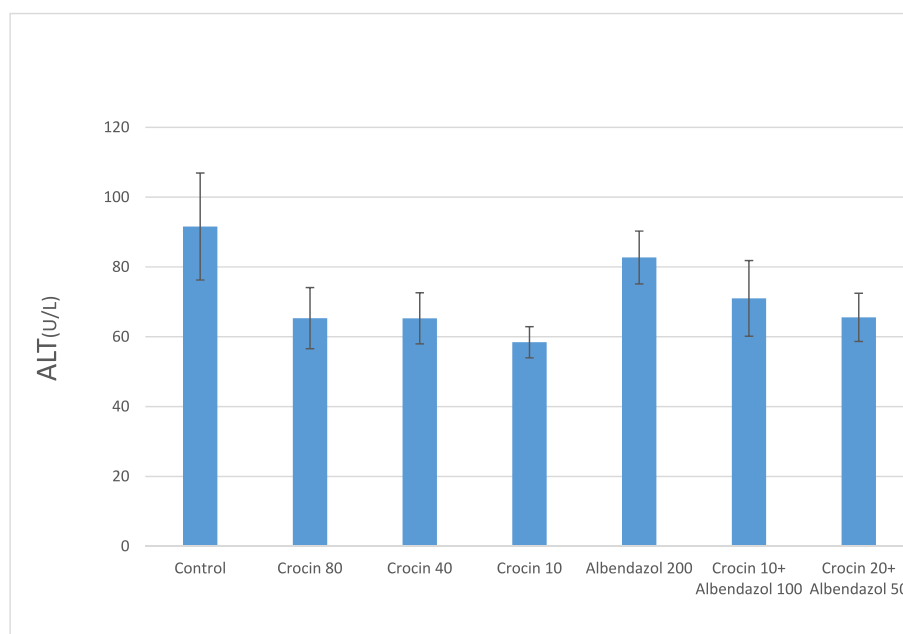


Fig. 7. Mean of ALT (IU/L) $\pm$ SD in both the treated and control groups.

mg/kg) compared to the control group; while, there was no significant difference for this enzyme between high doses of crocin (40, 80 mg/kg), albendazole and control groups. The observed decrease in ALT activity along with diminished bilirubin concentration in 10 mg/kg crocin group revealed that a low dose of crocin might have hepatoprotective effects in mice infected with cystic echinococcosis. It should be noted that increased ALT activity is one of the most sensitive indicators of liver injury. The significant decrease in plasma ALT levels by the crocin extract (10 mg/kg) suggests a decrease in hepatic injury in the investigated mice. In agreement with our findings, the application of 20 mg/kg *C. sativus* in Wistar rats with fatty liver had been able to decrease ALT (Mashmoul et al., 2016). Moreover, in another study (Yousefsani et al., 2018), crocin (50 and 100 μ g/ml) reduced liver toxicity and stopped liver damage related to oxidative stress. However, based on other studies (Ebrahimi et al., 2019), *C. sativus* could not considerably decrease liver enzymes as compared to control animals. Additional conclusive confirmation of the hepatoprotective or hepatotoxic properties of crocin is required.

About the toxic properties of saffron, it appears that doses of 1 g per day are required. Based on previous studies investigation regarded as the safety of oral crocin in mice and rats after treatment with crocin (up to 3 g, p.o. and i.p. as well as 15–180 mg/kg, i.p.), suggest that saffron and crocin in doses of milligrams per day can be inspected nontoxic during a one-month study (Mohamadpour et al., 2013; Modaghegh et al., 2008).²⁰ The results revealed that at pharmacological doses crocin did not cause damage to any major organ in the body.

Based on our results, the administration of crocin used at 10 mg/kg concentrations seems a hopeful applicant for the treatment of cystic echinococcosis. However, further studies are needed to investigate the immunological properties of crocin in the treatment of CE.

Author statement

The authors declared that there is no conflict of interest.

Declaration of competing interest

The authors declare there is no conflict of interest.

Data availability

No data was used for the research described in the article.

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